

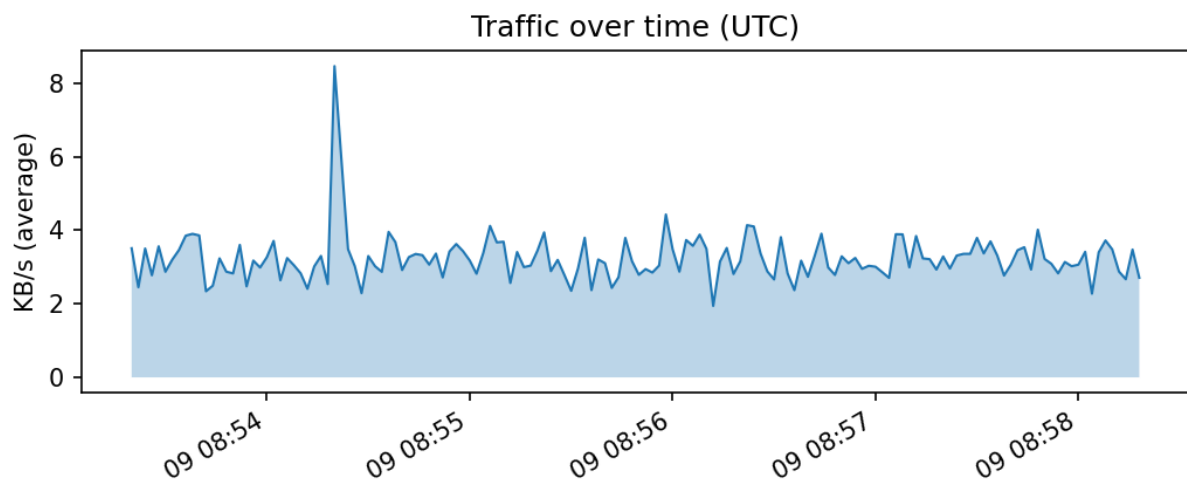
# Network Forensics Report

| Item                 | Value                                                              |
|----------------------|--------------------------------------------------------------------|
| File                 | sample.pcap                                                        |
| Size                 | 996.2 KB                                                           |
| SHA-256              | fff7938a45b0aa6e912c5770e27a0f0db5c65b948575deaedcc16be9e6379aed   |
| Packets analysed     | 1,558                                                              |
| Capture window (UTC) | 2025-10-09 08:53:20 to 2025-10-09 08:58:19 (300s)                  |
| Total data           | 971.8 KB                                                           |
| Unique IP hosts      | 8                                                                  |
| Tool                 | TShark (Wireshark) 4.2.2 (Git v4.2.2 packaged as 4.2.2-1.1build3). |
| Report generated     | 2026-10-03 23:50:33 UTC                                            |

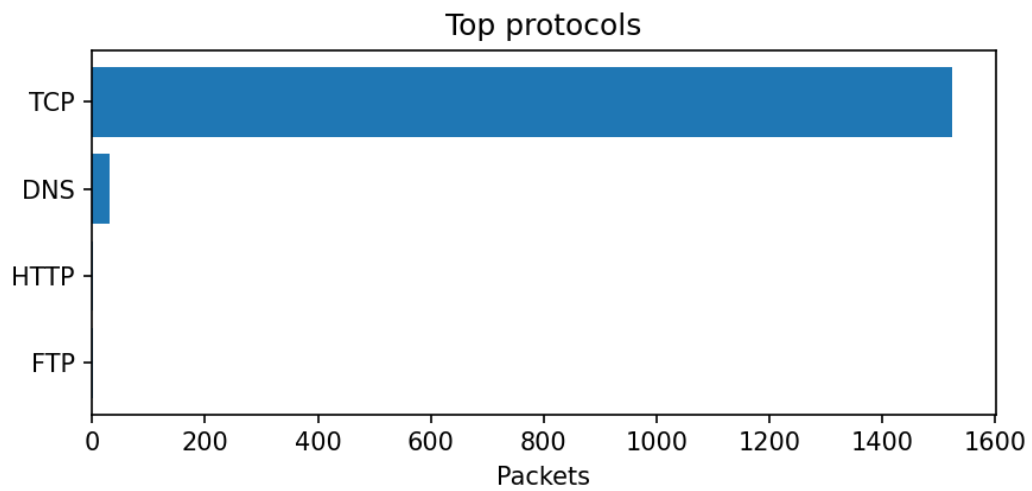
## Indicators (heuristics, not conclusions)

| Indicator          | Detail                                                                           |
|--------------------|----------------------------------------------------------------------------------|
| Possible port scan | 192.168.1.77 contacted 300 distinct destination ports                            |
| Possible beaconing | 192.168.1.20 -> 192.168.1.1:53 every ~9s (31 packets, very regular)              |
| Possible beaconing | 192.168.1.20 -> 198.51.100.7:8443 every ~10s (25 packets, very regular)          |
| Unusual DNS name   | aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa.bbbbbbbbbbbbbbbbbbbbbbb.tunnel.example.net |
| Cleartext protocol | FTP: 1 packets (not encrypted)                                                   |
| Cleartext protocol | HTTP: 1 packets (not encrypted)                                                  |

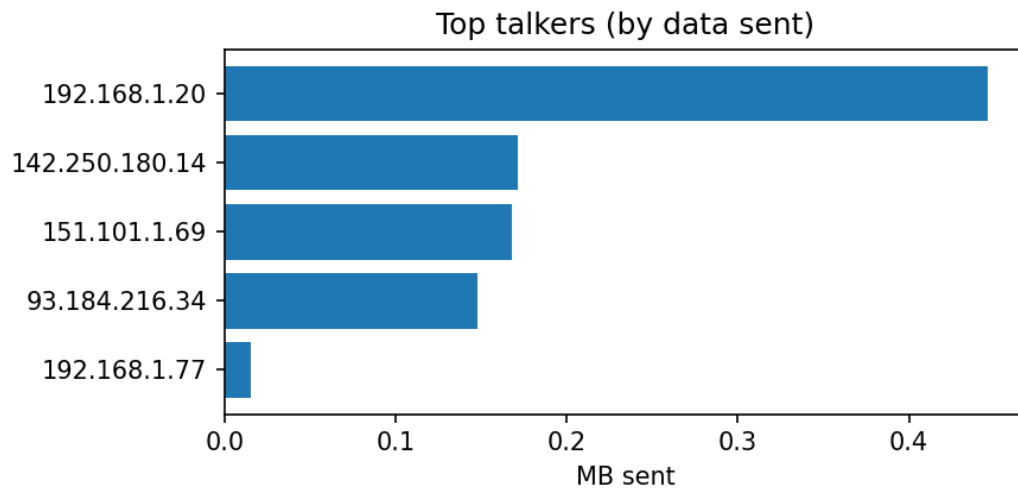
## Traffic timeline



## Protocols

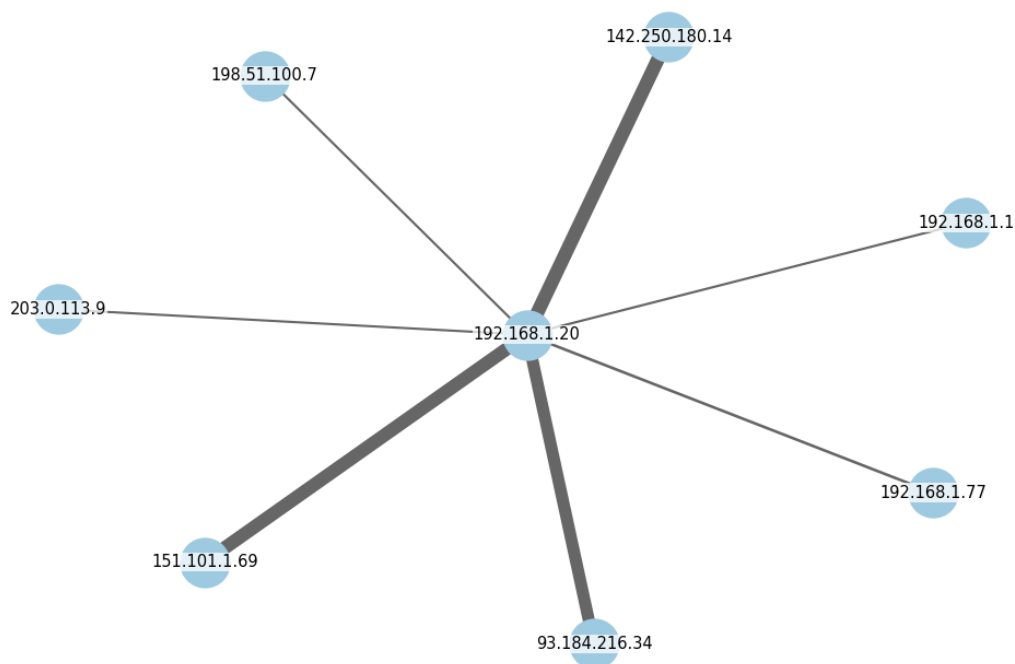


## Top talkers



## Conversation graph

Conversation graph (line thickness = data volume)



## Top conversations

| Host A         | Host B        | Packets | Data     |
|----------------|---------------|---------|----------|
| 142.250.180.14 | 192.168.1.20  | 424     | 336.3 KB |
| 151.101.1.69   | 192.168.1.20  | 408     | 325.0 KB |
| 192.168.1.20   | 93.184.216.34 | 369     | 290.9 KB |
| 192.168.1.20   | 192.168.1.77  | 300     | 15.8 KB  |
| 192.168.1.1    | 192.168.1.20  | 31      | 2.3 KB   |
| 192.168.1.20   | 198.51.100.7  | 25      | 1.4 KB   |
| 192.168.1.20   | 203.0.113.9   | 1       | 70 B     |

## DNS queries

| Name                                                                                 | Count |
|--------------------------------------------------------------------------------------|-------|
| example.com                                                                          | 10    |
| news.site.org                                                                        | 10    |
| cdn.fastly.net                                                                       | 10    |
| aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa.bbbbbbbbbbbbbbbbbbbbbbbbbbb.tunnel.example.net | 1     |

## HTTP hosts

| Name        | Count |
|-------------|-------|
| example.com | 1     |

## Methodology and limitations

- Packets were read with tshark and analysed with pandas; charts use matplotlib.
- Indicators are simple heuristics (distinct-port counts, interval regularity, DNS name length, cleartext protocols, traffic bursts). Legitimate traffic such as CDNs, games or keep-alives can trigger them.
- Encrypted traffic (TLS/HTTPS/QUIC) shows metadata only: addresses, names, sizes and timing.
- This is an automated triage aid, not a certified forensic examination.